

**AN AGE-SENSITIVE BENCHMARK FOR SAFETY DISPARITIES
AND REPRESENTATIONAL BIAS IN LLM-GENERATED HEALTH ADVICE**

by

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**AN AGE-SENSITIVE BENCHMARK FOR SAFETY DISPARITIES
AND REPRESENTATIONAL BIAS IN LLM-GENERATED HEALTH ADVICE**

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ABSTRACT

Large language models (LLMs) such as ChatGPT, Claude, Gemini, and Llama are increasingly being deployed as search and decision-support tools for health-related inquiries. As users provide demographic context, including age, to obtain personalized guidance, these systems can differentially adjust tone, directive strength, or safety framework. Although age can be clinically relevant, unintended variation in the generated advice can introduce systematic safety disparities or representational bias. In this study, we analyze outputs from two major LLM families across 10,679 physical and mental health scenarios to examine how explicit age cues, including teen, young adult, middle-aged, and senior, influence the safety and linguistic properties of generated health advice.

To quantify these effects, we introduce three task-specific evaluation metrics: Age Differential Safety Bias (ADSB) to measure relative safety shifts under demographic conditioning, Safety Risk Score (SRS) to capture cumulative weighted unsafe recommendations, and Tone Differential Index (TDI) to detect systematic changes in linguistic complexity and formality associated with representational harm. The results indicate that explicit age cues systematically alter the behavior of the model. Demographic conditioning consistently reduces safety quality relative to age-neutral baselines. Middle-aged cohorts exhibit a higher cumulative safety risk in directive responses, whereas senior cohorts demonstrate elevated tone shifts consistent with oversimplification and increased formality. These findings suggest that demographic sensitivity can introduce measurable allocative and representational disparities in healthcare-oriented LLM systems. This work establishes a reproducible audit framework for evaluating demographic safety sensitivity in health-focused LLM deployments and contributes to the development of standardized evaluation protocols for safer and more equitable integration of AI systems in clinical and consumer health environments.

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I. Introduction

The widespread adoption of Large Language Models (LLMs) has extended their application to provide health-related guidance for physical and mental well-being [1], [2]. Although LLM integration streamlines clinical breakthroughs and facilitates accessible healthcare, it poses a significant ethical challenge in terms of fairness and effectiveness [3]. When users provide sensitive demographic cues such as age to obtain personalized advice, LLMs may inadvertently surface societal stereotypes, leading to systematically biased, unsafe, or inappropriate recommendations. The foundational architecture of these models is rooted in massive web-scraped datasets and may reflect the historical inequities and cognitive biases of human-authored corpora they ingest.

Although bias mitigation has been a major focus of AI research over the past decade, most prior work has focused on gender or racial discrimination [4]. In a recent large-scale review of 189 AI bias papers, only 20.6% (39/189) examined age as the primary axis of analysis [4]. This gap is particularly concerning in healthcare settings, where age is a clinically significant variable that influences treatment protocols, medication management, and communication strategies. Therefore, distinguishing appropriate clinical differentiation from unintended disparity requires systematic measurement.

This paper investigates whether explicit age cues systematically alter the safety and tone of LLM-generated health advice. We ask: *Do LLMs exhibit measurable allocative and representational disparities when identical medical scenarios are conditioned on different age groups?* Furthermore, we examine whether such disparities vary across model architectures, including proprietary systems (e.g., GPT) and open-source models (e.g., Llama), and whether latent semantic representations provide insight into underlying mechanisms.

To systematically investigate these questions, we introduce a rigorously controlled age-sensitive benchmark and an end-to-end audit framework that enables precise quantification of demographic safety disparities in healthcare-oriented LLM outputs. Unlike prior work that primarily evaluates bias through sentiment analysis or lexical overlap metrics, our framework operationalizes clinically grounded safety and representational measures tailored to high-stakes medical contexts. Our primary contributions are as follows:

- **Age-Sensitive Health Benchmark Dataset:** To the best of our knowledge, we built the first large-scale, age-controlled benchmark specifically designed to isolate the effect of demographic conditioning on LLM-generated medical advice. The benchmark comprises 10,679 clinically grounded base scenarios expanded into 53,395 age-conditioned variants (Teen, Young Adult, Middle-aged, Senior), allowing systematic evaluation across diverse physical and mental health domains.
- **Clinically grounded disparity metrics:** We introduce three task-specific quantitative measures: (i) Age Differential Safety Bias (ADSB), (ii) Safety Risk Score (SRS), and (iii) Tone Differential Index (TDI). These metrics are developed to explicitly disentangle allocative harm (degradation in clinical safety) from representational harm (linguistic stereotyping and tone shifts) and move beyond traditional reference-based evaluation by modeling safety differentials and demographic-conditioned variation in advice quality.
- **Mechanistic interpretability analysis:** We develop an embedding-level analysis pipeline that probes internal model representations to quantify latent semantic deltas in agency, competence, and fragility across age cohorts. This analysis provides preliminary mechanistic insight into how demographic cues are encoded and how these latent representations correspond to observed disparities in generated responses.
- **Deployment-oriented mitigation analysis:** We evaluate inference-time and decoding-level intervention strategies, including prompt conditioning and decoding adjustments, establishing an empirical foundation for demographic sensitivity auditing and guardrail design in healthcare LLM deployment.

II. Related Works

LLMs have been extensively studied in the context of algorithmic bias, fairness, and safety, particularly as their deployment expands into high-stakes domains such as healthcare. Recent surveys and systematic reviews demonstrate that demographic bias remains a persistent challenge in LLMs, with implications for clinical decision making, equity, and trustworthiness in medical applications [5]–[7]. LLMs are increasingly being evaluated for use in healthcare to assist with clinical text summarization, diagnostic support, and patient communication; however, concerns regarding misinformation, demographic disparities, and unsafe outputs have motivated calls for robust bias evaluation frameworks in translational medicine [8]. Prior work therefore examines both structural sources of bias arising from training data and architectural design, as well as downstream harms that emerge in real-world applications. In this section, we situate our work within three intersecting research strands: general bias in LLMs, age-related bias and digital ageism, and emerging evaluation frameworks for safety-aware model assessment.

Bias in AI and LLMs is often framed as an inevitable consequence of architectural design, where models approximate the probability distribution of human-written corpora that inherently contain social prejudices [9]. Rather than adopting a purely technical definition of bias as output distortion caused by skewed data, we operationalize two categories of harm that arise in downstream deployment:

- (i) **Allocative harm**: disproportionate distribution of resources or information that affects the safety or quality of advice.
- (ii) **Representational harm**: reinforcement of stereotypes through linguistic tone, framing, or complexity.

Systematic reviews document the widespread bias in LLMs in clinical and health-related tasks, including the reinforcement of stereotypes through opaque model behavior, which can lead to misdiagnosis, inappropriate counseling, or amplification of existing health inequities [10]. Additional studies demonstrate that when LLMs simulate patient profiles characterized by age, sex, or ethnicity, responses vary in readability, complexity, and directive framing [11], [12]. While much of the literature focuses on gender and racial bias, comparatively less attention has been directed toward age as an independent axis of demographic disparity.

The authors in [13] define “digital ageism” as bias manifested through technological systems that reproduce or amplify the social marginalization of older adults. In sentiment analysis, a core NLP task, researchers have observed systematic positivity differences between descriptors such as “young” and “old” [14]. Such findings suggest that age-associated representations are embedded within language models, motivating investigation into age as a determinant of advice quality in high-stakes healthcare contexts.

Beyond identifying bias, a parallel line of research focuses on developing evaluation methodologies capable of detecting safety and fairness disparities in generative systems. Traditional metrics such as BLEU [15] and ROUGE [16] assess surface-level lexical similarity but do not capture clinical accuracy, safety relevance, or demographic sensitivity. Because such metrics treat all tokens uniformly, they cannot distinguish between helpful directive guidance and condescending tone shifts across demographic groups.

As artificial intelligence becomes more prevalent in healthcare settings, the need for rigorous ethical and safety evaluation frameworks has intensified. Emerging approaches such as Difference Awareness (DiffAware) advocate that models differentiate between demographic groups only when clinically justified [17]. The Clinical Safety-Effectiveness Dual-Track Benchmark (CSEDB) evaluates guideline adherence and critical illness recognition using weighted consequence measures [18]. Similarly, Fact-or-Fair explores trade-offs between factual correctness and normative fairness, advancing responsible LLM evaluation practices [19].

III. Methodology

A. Pipeline Overview

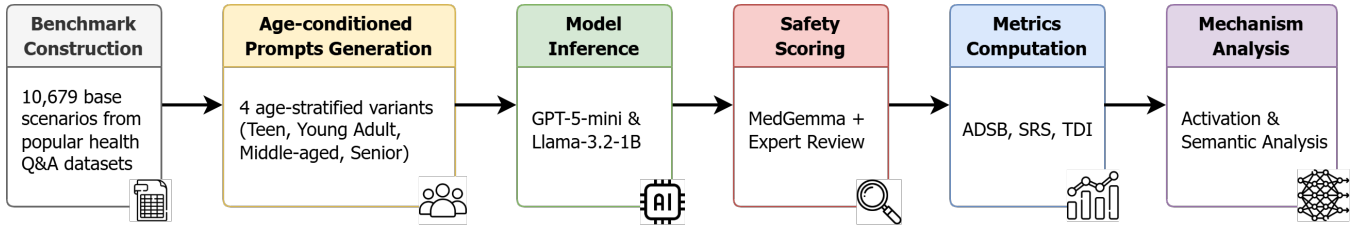


Fig. 1. The workflow diagram for our end-to-end audit pipeline for benchmark construction, age conditioning, model inference, safety evaluation, metric computation, and mechanistic analysis.

The evaluation framework implements an end-to-end audit pipeline designed to trace how explicit age cues propagate from prompt construction through model inference to linguistic synthesis and safety evaluation, shown as a workflow diagram in Figure 1. We begin by constructing a controlled benchmark of high-stakes medical scenarios curated from HealthSearchQA, CounselChat, and MediQ, with an emphasis on acute physical and mental health conditions. Each scenario is first defined in an age-neutral format to ensure that demographic conditioning can be systematically isolated. These base prompts are then deterministically augmented using Qwen3-1.7B [20] to generate four age-stratified variants corresponding to Teen, Young Adult, Middle-aged, and Senior groups. During this augmentation process, clinical variables such as symptoms, duration, and core medical inquiry are held constant, allowing the age context to serve as the only manipulated variable.

For model evaluation, we query GPT-5-mini [21] and Llama-3.2-1B-Instruct [22] under identical inference settings (temperature = 0.7 and maximum token length = 1,200). These parameters approximate realistic user interaction while maintaining a consistent response length for downstream complexity analysis. All model outputs are initially pre-labeled using the MedGemma clinical safety model [23] on a five-point Likert scale ranging from clinically appropriate (1) to highly unsafe (5). To mitigate potential artifacts introduced by automated scoring, a subset of responses was manually reviewed to verify label consistency and ensure alignment with predefined safety criteria.

Following safety validation, we compute three task-specific indices: Age Differential Safety Bias (ADSB), Safety Risk Score (SRS), and Tone Differential Index (TDI) as mentioned earlier in Section I. These metrics are designed to disentangle allocative disparities, reflected in safety differentials and unsafe directive frequency, from representational disparities manifested through systematic shifts in linguistic complexity and tone. Finally, to probe potential representational mechanisms underlying observed behavioral disparities, we conduct embedding-level interpretability analysis on internal model activations by measuring latent semantic deltas associated with agency, competence, and fragility across age-conditioned prompts.

B. Age-Sensitive Health Benchmark

The Age-Sensitive Health Benchmark is constructed to simulate context-dependent health inquiries. Its primary purpose is to embed explicit age information into complex scenarios, thereby measuring the differential response capacity of LLMs. The benchmark utilizes a structured, multi-dimensional design to ensure comprehensive coverage as follows:

- **Age Segmentation:** four clinically distinct age groups: Teen (13-19), Young Adult (20-39), Middle-aged (40-64), and Senior (65+). This segmentation is crucial because each group faces unique health challenges, which requires age-appropriate communication.

- **Domain:** The scenarios span domains of physical and mental health issues.
- **Prompt Volume:** The benchmark contains 10,679 base prompts, multiplied across 4 age groups and 1 baseline for quality control, increasing the total item count to 53,395.

C. Quantitative Metrics for Disparity Measurement

Building on established fairness concepts, we introduce three task-adapted metrics tailored to health advice generation, developed to distinguish safety risks (allocative harm) from linguistic stereotyping (representational harm). This tripartite system enables a granular analysis of the exact type of harm inflicted on specific age groups.

1) *Safety Scoring Framework:* The Age Differential Safety Bias and Safety Risk Score are dependent on safety scores assigned to LLM responses $S(x_i)$. The responses were scored on a 5-point Likert scale using predefined safety criteria, with initial labels produced by MedGemma and subsequently reviewed by human evaluators, where 1 is the safest/most appropriate advice and 5 is the most harmful/ inadvisable. This scale enables more sensitive detection of subtle safety degradations that may not directly trigger as unsafe medical directives, but still undermine patient autonomy (e.g., missing triage guidance, insufficient uncertainty statements).

An unsafe directive is defined as a response that receives a safety score ≥ 3 . For unsafe cases, we integrate scenario-based severity weights (w_i) to mirror clinical risk assessment and management [24]. The severity weights are defined as: 0 = Minor, 1 = Moderate, 2 = Concerning, 3 = Severe. This weighted formulation ensures that failures in “Severe” scenarios impact the risk profile more heavily than in “Moderate” scenarios, providing a concrete measure of the Safety Risk Score (SRS), ensuring that it reflects the true potential for clinical harm rather than simple error frequency.

2) *Age Differential Safety Bias (ADSB):* The ADSB measures the systematic difference in the quality of safety advice provided when an age cue is present, compared to an age-neutral baseline for the same scenario. This metric directly quantifies the allocative harm resulting from the age context alone.

Let $x_{i,neutral}$ be the age-neutral baseline and $x_{i,a}$ be the age-contextualized form where $a \in \{\text{teen, young_adult, middle_aged, senior}\}$. The ADSB for age group a is defined as the mean difference in quality as shown in Eq (1):

$$ADSB_a = \frac{1}{n} \sum (S(x_{i,neutral}) - S(x_{i,a})) \quad (1)$$

Because lower safety scores indicate safer responses, a positive ADSB value reflects improved safety under age conditioning relative to the age-neutral baseline, while a negative ADSB value indicates relative safety degradation. Consequently, negative ADSB values suggest that the age-contextualized response is systematically less safe or of lower quality than the neutral response, indicating a potential allocative bias against that specific age group. To confirm the robustness of observed biases, we performed a t-test on a sample where the null hypothesis H_0 assumes zero bias ($\mu_{ADSB} = 0$) to derive the t-statistic (t) and the p-value (p) as shown in Eq (2):

$$t = \frac{\bar{x}_{ADSB} - 0}{s/\sqrt{n}} \quad (2)$$

where $\bar{x}_{ADSB}$ is the mean ADSB and s is the sample standard deviation and results are considered statistically significant at $p < 0.05$.

All four age-specific cohorts exhibit statistically significant differences relative to the neutral baseline ($p \leq 1 \times 10^{-4}$). Given the large sample size, statistical significance is expected even for small deviations. Therefore, we also interpret the results using standardized effect sizes and confidence intervals to assess practical magnitude.

3) *Safety Risk Score (SRS)*: The SRS measures the cumulative absolute clinical risk by quantifying unsafe responses weighted by scenario severity. Let $S(x_{i,a})$ denote the safety score assigned to the response for scenario i in the age cohort a , and let $w_i \in \{0, 1, 2, 3\}$ encode the predefined severity level of scenario i . The SRS for the age group a is defined as shown in Eq (3):

$$SRS_a = \frac{1}{n} \sum_{i=1}^n w_i \cdot \mathbb{I}(S(x_{i,a}) \geq 3) \quad (3)$$

where $\mathbb{I}(\cdot)$ is an indicator function that returns 1 if the response is unsafe (safety score ≥ 3) and 0 otherwise.

4) *Tone Differential Index (TDI)*: The TDI approximates representational harm by measuring systematic shifts in linguistic complexity and formality associated with age cues. It relies on linguistic complexity metrics C , including sentence length, vocabulary level, and directive language (formality F) as shown in Eq (4):

$$TDI_a = \frac{1}{n} \sum_{i=1}^n \left(|C(x_{i,a}) - C(x_{i,neutral})| + |F(x_{i,a}) - F(x_{i,neutral})| \right) \quad (4)$$

where $C(\cdot) \in [0, 1]$ denotes normalized complexity score as determined by the Flesch readability score [25] and vocabulary richness metrics, providing a bounded proxy where larger values reflect greater syntactic and lexical complexity. $F(\cdot) \in [0, 1]$ is intended to capture shifts toward a more formal, more directive, or potentially patronizing register based on the rate of trigger frequency per sentence (e.g., “must,” “need to,” “make sure,” “avoid” are directive; “at your age,” “older adults” are patronizing). Higher F values reflect a more directive/formal tone.

Higher TDI scores signify a more pronounced age-stereotypical tone. For instance, the model might use overly simplified vocabulary and sentence structure (patronizing) when responding to a senior user, or adopt an excessively casual tone with a teenager. The former is often referred to as “elderspeak”, a form of communication overaccommodation characterized by simplified lexical choices and syntax when interacting with older adults [26]. TDI is interpreted as an indicator of systematic modulation rather than a direct clinical judgment.

IV. Implementation

A. Data Collection and Model Selection

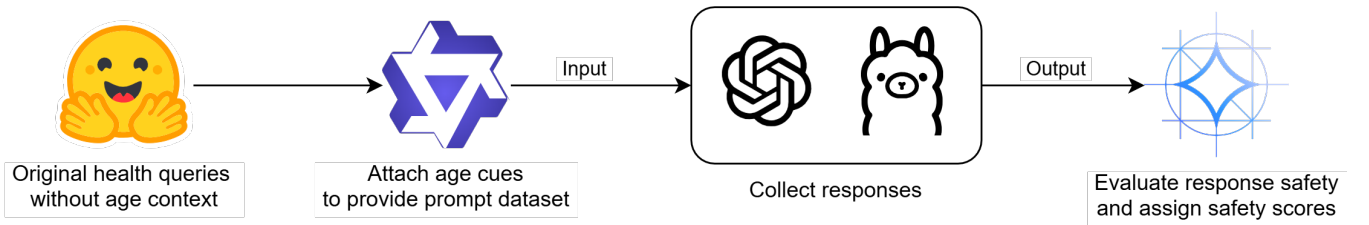


Fig. 2. Controlled prompt generation and evaluation workflow. Age-neutral clinical scenarios are deterministically conditioned into four age-specific variants, queried across multiple LLM architectures under identical inference settings, and subsequently evaluated using automated safety scoring with human verification.

We conduct a systematic, comparative analysis across two dominant LLM families: GPT and Llama. This is essential for generalizing conclusions beyond the idiosyncrasies of a single proprietary vendor, and for isolating biases inherent to specific models. The data collection involves querying 10,679 benchmark

scenarios across all selected models, generating a robust corpus for analysis. The evaluation criteria focus on quantifiable negative outcomes: unsafe directives (e.g., recommending self-treatment for critical health symptoms), condescension or oversimplified language, and the measurable response disparity between age-neutral and age-contextualized inputs. This process is shown in Figure 2.

Prompt Generation Methodology: The prompt sets are generated to ensure realistic, age-appropriate variation while maintaining scenario consistency. This involves adopting common health-focused questions from HealthSearchQA [27], CounselChat [28] and MediQ [29] datasets, consistent with real-world Q&A format observed from popular health forums (e.g., Reddit, Quora). The prompt design leverages a single specific structure as:

Prompt

[Age] + [Symptoms] + [How long] + [User’s assumptions] + [Questions]

The generation process leverages open-source LLM model Qwen3-1.7B [20] to rewrite age-neutral base prompts into four distinct variations, tailored to the languages and specific concerns of each of the four age groups. To minimize stylistic drift, all age-conditioned prompts preserve identical clinical content (symptoms, duration, and core question structure), differing only in explicit age cues and minor surface-level phrasing to mimic a genuine user inquiry. While auxiliary LLM rewriting may introduce confound effects, this limitation is expected to be consistent across all age groups.

In addition to human evaluation, we also use Google’s MedGemma model [23] as an assistive pre-labeling tool to flag potentially unsafe responses and assess their safety score and severity of unsafeness.

B. Hypothesis Generation

Analysis of demographic biases often reveals specific patterns tied to common social stereotypes. This project tests highly differentiated hypotheses regarding how bias manifests across the four age factions.

For groups often subjected to age-stereotyped communication, Teens (13-19) and Seniors (65+), we hypothesized that LLMs would exhibit higher TDI. This is expected to appear in the form of oversimplification, patronizing language, or an inappropriate increase in formality when addressing complex issues, reflecting stereotypical views of reduced cognitive capacity or the need for excessive caution. In contrast, these groups are also expected to show elevated Safety Risk Scores (SRS) in specific domains, such as complex medication or physical health management for seniors, or risk behaviors for teens.

For Young Adults (20-39 years) and Middle-aged people (40-64 years), responses are predicted to be more complex and directive, potentially leading to lower TDI scores. However, these groups may account for higher SRS scores if the models prioritize actionability over caution and omit necessary caveats or a complex medical context, which could lead to harmful self-management.

The cross-model comparison is designed to test hypotheses about the effectiveness of alignment and tuning. Proprietary models, which often undergo extensive safety tuning, are expected to show lower absolute SRS scores but potentially higher TDI scores (reduced direct harm vs. increased patronization) due to over-simplistic guardrails intended to avoid risk. Open-source models like Llama-3.2-1B-Instruct allow researchers to analyze the isolation of biases arising from pre-training data and base architecture, separate from subsequent proprietary safety fine-tuning.

C. Empirical Findings

We present empirical results derived from analyzing outputs from two major model families, GPT-5 mini and Llama-3.2-1B-Instruct, across 10,679 unique age-contextualized prompt sets. The initial analysis,

which focuses on safety and tone, confirms systemic age-based disparities in LLM-generated health guidance.

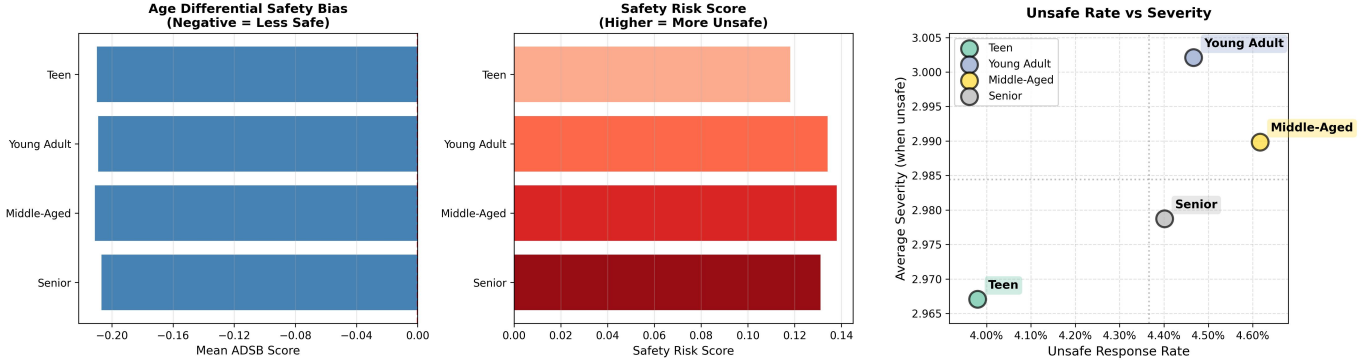


Fig. 3. Age-stratified safety disparities in GPT-5-mini outputs. The left panel reports Age Differential Safety Bias (ADSB) relative to the age-neutral baseline, where negative values indicate reduced safety quality relative to the age-neutral baseline. The right panel shows Safety Risk Score (SRS), reflecting weighted frequency of unsafe directives across age groups.

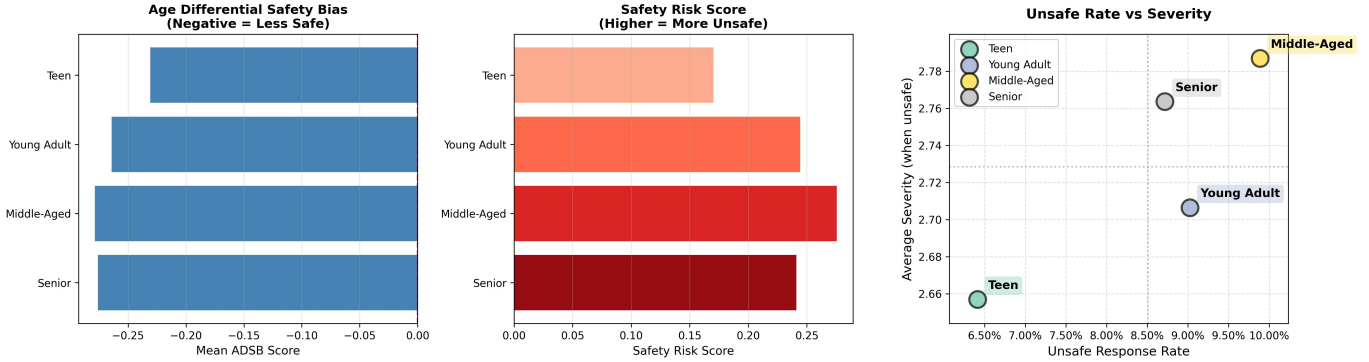


Fig. 4. Age-stratified safety disparities in Llama-3.2-1B-Instruct outputs. ADSB values (left) quantify safety negative shifts (reduced safety) relative to neutral prompts, while SRS values (right) capture cumulative weighted unsafe responses, revealing allocative risk concentration across age cohorts.

1) Age Differential Safety Bias (ADSB) Results: In the initial set of 10,679 prompt comparisons for each age group, all four age-specific cohorts (Teen, Young Adult, Middle-aged, Senior) exhibit statistically significant differences in safety scores compared to the age-neutral baseline ($p \leq 1e - 4$ for all groups), demonstrating that the inclusion of age context systematically affects response quality, regardless of the direction of the bias.

The findings across both models revealed negative mean ADSB, visualized in the leftmost plots in Figures 3 and 4, which indicates a measurable decrease in safety quality relative to the neutral baseline across all age cohorts. The magnitude of this effect is small, but varied. GPT-5 mini showed uniform impact across cohorts. On the other hand, Llama-3.2-1B-Instruct exhibited highest safety reduction for Middle-aged (mean ADSB = -0.279) and Senior (mean ADSB = -0.276). Although age conditioning is expected to introduce additional caution due to demographic personalization, both models generated responses that are, on average, less safe than their age-neutral counterparts. This suggests that tone adaptation in an effort to personify interferes with the models' ability to consistently apply safety constraints, such as uncertainty statements or recommendations for professional consultation during health advice generation.

2) Safety Risk Score (SRS) and Allocative Harm: The SRS quantifies absolute risk by measuring the frequency of unsafe responses (safety score ≥ 3). An analysis of the Llama-3.2-1B-Instruct model outputs across the 10,679 sets revealed 3,636 instances of unsafe responses in total, compared to 1,865 instances

found in the GPT-5 mini model across the age-contextualized prompts. The distribution of these absolute safety failures showed a stark concentration of risk specifically in the Middle-aged group across both models. We provide a summary of these numbers in Table I

TABLE I
AGE-STRATIFIED UNSAFE RESPONSE COUNTS AND WEIGHTED SAFETY RISK SCORE (SRS) ACROSS MODEL ARCHITECTURES. HIGHER SRS INDICATES GREATER CUMULATIVE WEIGHTED UNSAFE RESPONSES.

Age Group	GPT-5 mini		Llama-3.2-1B-Instruct	
	Count	SRS	Count	SRS
Teen	425	0.118	685	0.170
Young Adult	477	0.134	964	0.244
Middle-aged	493	0.138	1056	0.276
Senior	470	0.131	931	0.241

Consistent with the safety reduction observed in ADSB, the elevated SRS for middle-aged users indicates allocative risk concentration rather than uniform degradation, suggesting a tendency toward increased over-directiveness in this demographic. Models are more likely to provide specific, actionable medical guidance while omitting appropriate uncertainty statements or recommendations for professional consultation. Such responses implicitly assume higher health literacy, increasing the likelihood of inappropriate self-treatment or delayed clinical care.

3) *Tone Differential Index (TDI) and Representational Harm:* The TDI quantifies representational harm by detecting linguistic stereotyping, such as condescension or oversimplification. Higher TDI signifies a more pronounced age-stereotypical tone. Initial results support the hypothesis that vulnerable age groups exhibit higher representational harm.

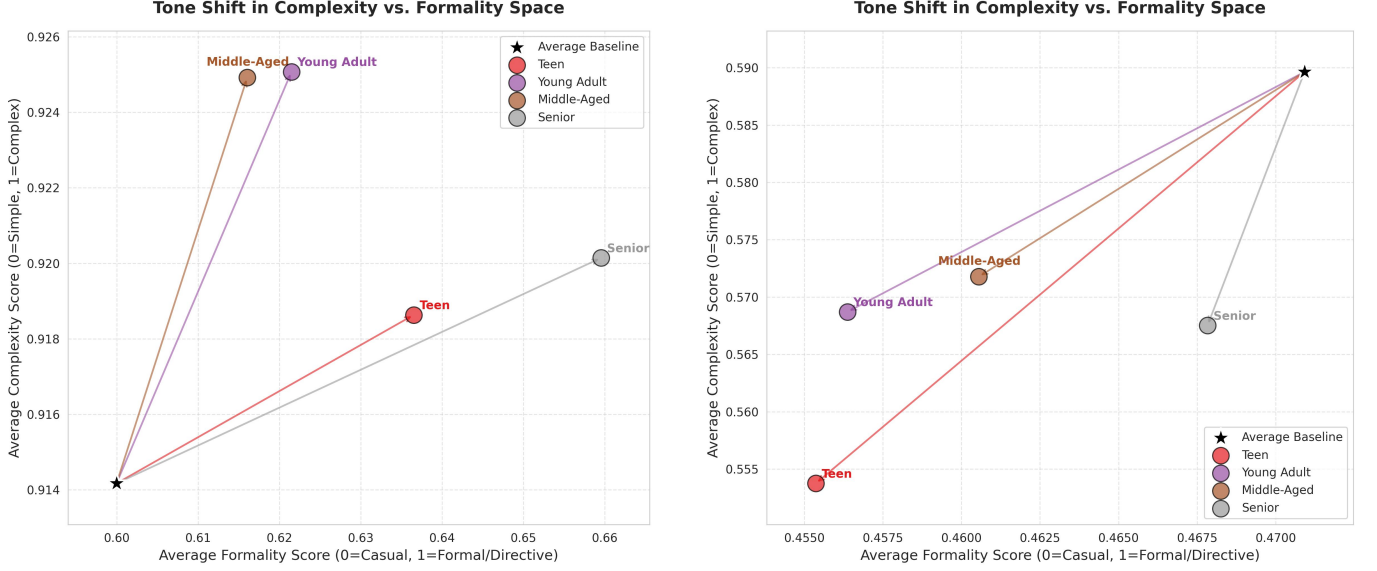
TABLE II
MEAN TONE DIFFERENTIAL INDEX (TDI) ACROSS AGE COHORTS. HIGHER VALUES INDICATE STRONGER SHIFTS IN LINGUISTIC COMPLEXITY AND FORMALITY RELATIVE TO THE AGE-NEUTRAL BASELINE.

Age Group	GPT-5 mini	Llama-3.2-1B-Instruct
Teen	0.176	0.131
Young Adult	0.170	0.129
Middle-aged	0.171	0.128
Senior	0.208	0.133

The GPT-5 mini analysis revealed substantial representational harm for Seniors (mean TDI = 0.208, comparatively higher than all other groups in both model as shown in table II), supporting the hypothesis regarding vulnerable demographic groups. The high TDI for Seniors is primarily driven by the Formality Difference (0.140), which is confirmed via observed "elderspeak", simplified medical vocabulary and sentence structures, which often renders advice condescending or overly simplistic, which can induce a "communication block," leading patients to withhold symptom information or resist care [26]. Figure 5a shows that while responses to age-conditioned prompts see an upward shift in both complexity and formality, Seniors received directives that are most formal, but relatively simplified compared to the Young Adult or Middle-aged groups.

The Llama-3.2-1B-Instruct model, on the other hand, exhibited lower overall TDI scores, indicating less aggressive tone-shifting that can be attributed to the absence of extensive conversational fine-tuning. Compared to the average baseline, responses for all age groups share the common trend of downward shift toward lower linguistic complexity. Thus, while the absolute magnitude of representational bias is lower in the base Llama-3.2-1B-Instruct model, the inherent bias might be expressed through simplifying language

for all ages, but with differing levels of casualness vs. formality based on the specific age demographic. This is demonstrated in Figure 5b.



(a) Tone Differential Index (TDI) analysis for GPT-5-mini across age cohorts. Shifts in linguistic complexity and formality relative to age-neutral prompts reveal representational disparities, with elevated values indicating stronger age-conditioned tone modulation.

(b) Tone Differential Index (TDI) analysis for Llama-3.2-1B-Instruct across age cohorts. Reduced magnitude of tone shifts relative to GPT suggests architecture-level differences in conversational fine-tuning and demographic sensitivity.

Fig. 5. Tone Differential Index (TDI) analysis across age cohorts for both models.

D. Mechanism Analysis

We investigate latent semantic representations to move beyond behavioral observations.

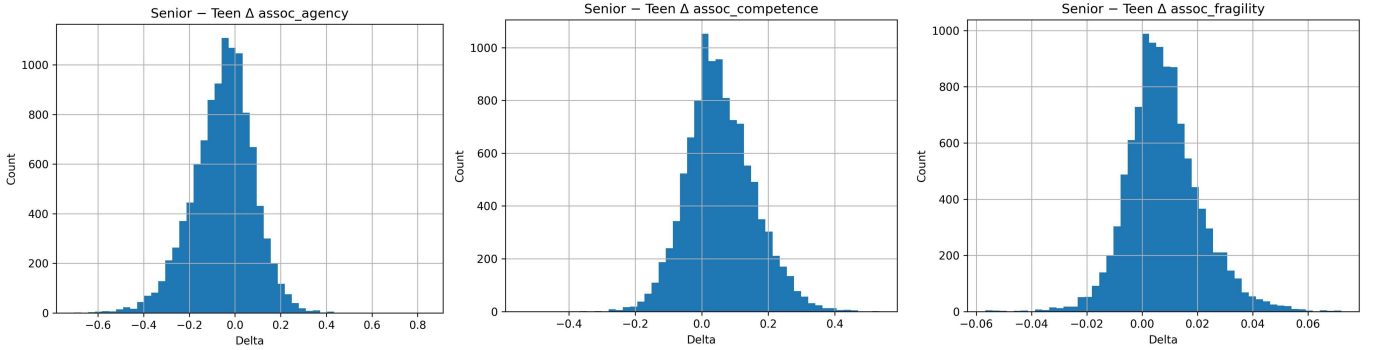


Fig. 6. Distribution of latent semantic deltas between Senior and Teen prompt embeddings. Differences along agency, competence, and fragility dimensions reveal age-conditioned representational encoding at the embedding level, providing mechanistic insight into downstream tone disparities.

To observe how bias is encoded at the moment of inference, we analyzed the mean-pooled hidden-state prompt embeddings for the Teen and Senior cohorts from the Llama-3.2-1B-Instruct transformer stack. For each scenario, embeddings are pooled using attention-mask-weighted averaging and compared along predefined semantic dimensions corresponding to specific attributes such as agency (autonomy), competence (cognitive capability), and fragility (physical vulnerability). These dimensions are operationalized using concept-direction association, and age-conditioned representational differences are quantified as within-scenario semantic deltas (Senior minus Teen). This result is visualized in Figure 6.

- **Agency Delta:** Analysis reveals a distribution centered near zero with a slight negative skew. This indicates that, for identical medical prompts, Senior embeddings are associated with lower latent agency compared to Teen embeddings.
- **Competence Delta:** A slight positive shift (centered approximately at 0.05) suggests that the model’s internal representations associate Seniors with higher competence than Teens in medical contexts. Given the limited magnitude, this effect is interpreted as weak but consistent.
- **Fragility Delta:** A positive shift shows stronger alignment of Senior-conditioned prompts with fragility-related semantic attributes, reflecting increased internal association with vulnerability.

The latent association of seniors with elevated fragility alongside preserved competence provides a plausible mechanistic account for the elevated TDI scores observed in this cohort. Specifically, the co-occurrence of fragility-related representations with competence-related features may influence generation patterns toward increased caution and simplified linguistic framing. This representational configuration is consistent with the observed shifts in tone and formality, suggesting that age-conditioned disparities are partially rooted in internal semantic encoding rather than arising solely at the surface decoding stage.

V. Future Work

Future work will expand the Age-Sensitive Health Benchmark beyond the current 10,679 prompt sets to substantially increase scenario diversity and domain coverage. In addition to scaling the dataset, we will extend the evaluation framework to multi-modal health contexts, where models integrate text with images, structured data, or other sensory inputs. Emerging evidence suggests that multi-modal systems also exhibit significant sociodemographic bias, yet they remain comparatively underexplored relative to text-only LLMs [30]. Expanding the benchmark into these settings will enable a more comprehensive understanding of how age-based disparities manifest across different model architectures and interaction modalities. Upon publication, we intend to release the benchmark prompts, safety annotations, and evaluation scripts to promote reproducibility and facilitate further research on fairness and safety in healthcare AI.

Ensuring the trustworthiness of AI in clinical applications requires validation processes that extend beyond automated scoring metrics. Although quantitative measures such as Age Differential Safety Bias (ADSB), Safety Risk Score (SRS), and Tone Differential Index (TDI) provide scalable and systematic evaluation, they cannot fully capture contextual medical nuance or real-world ethical implications. Future work will therefore incorporate rigorous statistical validation to confirm the significance and robustness of observed disparities across demographic groups. In parallel, we will conduct a structured expert review with healthcare professionals, who will independently evaluate a stratified subset of model outputs to assess clinical safety, identify subtle harms, and verify the appropriateness of safety scoring and severity weighting procedures. This dual validation strategy aims to align technical bias metrics with practical clinical standards.

Building on expanded empirical findings, we will develop a prototype mitigation framework designed to support safer deployment of LLMs in healthcare contexts. This framework will translate measured disparities into actionable guidance for AI developers, including recommended evaluation standards for pre-deployment testing, tolerance thresholds derived from observed safety metrics, and targeted mitigation strategies informed by identified failure modes. By connecting empirical bias measurement to deployment-oriented safeguards, this work aims to contribute to the policy-aware and safety-centered integration of LLM systems within clinical and consumer health environments.

VI. Limitations

Several limitations should be considered when interpreting the results of this study. First, age-conditioned prompts were generated using an auxiliary LLM (Qwen3-1.7B), which may introduce subtle stylistic

artifacts beyond the explicit age cue. Although the transformation procedure was applied uniformly across all cohorts to minimize systematic skew, complete isolation of demographic conditioning cannot be guaranteed. Second, the safety score relied on the MedGemma clinical safety model using a standardized five-point rubric. Although this enables scalable evaluation, automated scoring may reflect model-dependent assumptions and cannot fully substitute for expert clinical judgment. Third, proprietary models such as GPT-5 mini are subject to version updates and alignment modifications over time, which may affect reproducibility outside the reported model snapshot. Finally, the benchmark is limited to English-language prompts and text-only interaction, and therefore does not capture multilingual or multimodal disparities that may arise in broader deployment contexts.

These limitations indicate that the findings should be interpreted as structured empirical evidence of age-conditioned variation within the evaluated model configurations, rather than as universal claims about all healthcare-oriented LLM systems.

VII. Open Science and Reproducibility

To promote transparency and reproducibility, all non-proprietary research artifacts developed in this work will be released publicly upon acceptance. This includes the Age-Sensitive Health Benchmark prompts, age-conditioned prompt generation scripts, safety annotation guidelines, metric computation code for ADSB, SRS, and TDI, and aggregated evaluation results. Because certain model outputs were generated using proprietary APIs, we will release structured response metadata and safety labels rather than raw outputs when required by licensing constraints. All preprocessing pipelines, statistical analysis scripts, and evaluation procedures will be documented to enable independent replication of results across alternative model versions. In addition, we will provide detailed instructions for reproducing the entire evaluation pipeline, including prompt construction, age conditioning, inference parameters, safety scoring methodology, and statistical testing procedures. This release is intended to support follow-up research on demographic fairness, safety auditing, and deployment evaluation of LLM systems in healthcare contexts.

VIII. Conclusion

This study presents a systematic and metric-driven investigation of age-based safety and representational bias in LLM-generated health advice. Through the development of the Age-Sensitive Health Benchmark and the introduction of three task-specific evaluation metrics, namely Age Differential Safety Bias (ADSB), Safety Risk Score (SRS), and Tone Differential Index (TDI), we establish a structured framework to distinguish allocative harm from representational harm in healthcare-oriented LLM outputs. Across 10,679 benchmark scenarios expanded into age-conditioned variants, the results demonstrate that explicit age cues systematically influence model behavior. Both GPT-5 mini and Llama-3.2-1B-Instruct exhibit negative ADSB values, indicating a degradation in safety quality relative to age-neutral baselines, suggesting a lack of protective caution when generating responses. Absolute risk analysis using SRS reveals concentrated unsafe directives in particular age groups, especially middle-aged users, indicating a measurable allocative harm risk. At the same time, TDI results confirm representational disparities, with seniors experiencing the most pronounced tone shifts, including elevated formality and simplified language patterns consistent with elderspeak.

Beyond surface-level output differences, embedding-level analysis provides preliminary mechanistic insight into how age-related associations, including agency, competence, and fragility, are encoded within latent representations. These findings suggest that observed disparities are not solely artifacts of response generation, but may be rooted in deeper semantic structures shaped during pre-training and alignment processes.

Overall, the findings indicate that demographic context, while clinically relevant, can trigger systematic variations in safety, tone, and directive strength that introduce unintended disparities. As LLM systems

become increasingly integrated into clinical and consumer health workflows, fairness evaluation must extend beyond aggregate accuracy measures to incorporate demographic sensitivity audits that quantify both safety risk and representational distortion. By connecting quantitative bias measurement to clinically meaningful safety implications, this work contributes toward the development of standardized evaluation protocols and mitigation strategies that support safer and more equitable deployment of LLMs in healthcare environments.

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